Fundamentos de Programación II

Descripción Ver entrega

Límite de entrega: viernes, 25 de abril de 2025, 10:30
Ficheros requeridos: prueba.py, tasks.py (Descargar)
Tipo de trabajo: Individual

Registro de Tareas Pendientes

Se pide completar el desarrollo de la clase `TaskList` para gestionar una lista de tareas pendientes. Cada tarea está representada por un objeto de la clase `Task` que se suministra, la cual dispone de dos atributos: `name` (una cadena de texto) y `duration` (un entero que representa el tiempo estimado en minutos que requiere la tarea). Para facilitar el mostrado de la lista, la clase `TaskList` hereda de la clase BaseContainer permitiendo el mostrado del contenido de la lista.

Las tareas en los objetos `TaskList` se almacenarán internamente mediante una **lista enlazada** (no se permite el uso de listas nativas de Python). Se proporciona la clase interna `Nodo` que define los elementos de esta lista enlazada. Cada nodo almacena una tarea.

Debes completar el código de la clase `TaskList` añadiendo los atributos y métodos necesarios para que cumpla con las siguientes funcionalidades:

Métodos y propiedades a desarrollar:

1. **Método** `add_task`
   Añade una nueva tarea pasada por parámetro al final de la lista.

2. **Método** `get_task`
   Devuelve la tarea que se encuentra en la posición pasada por parámetro (comenzando desde 0). Si la posición no es válida, devuelve `None`.

3. **Método** `task_completed`
   Elimina y devuelve la tarea en la posición dada como parámetro, si existe. Esta acción debe considerar la propiedad `time_expent`. Devuelve `None` si la posición es incorrecta.

4. **Propiedad de solo lectura** `time_left`
   Devuelve el tiempo total (en minutos) de todas las tareas **pendientes** en la lista.

5. **Propiedad de solo lectura** `time_spent`
   Devuelve el tiempo total que se ha dedicado a las tareas **completadas**.

Recomendaciones:

- No se permite el uso de listas nativas para almacenar las tareas.
- Asegúrate de manejar correctamente los índices fuera de rango.
- Puedes usar un script de prueba (por ejemplo, `prueba.py`) para verificar el funcionamiento de tu implementación, su contenido no será evaluado.
- El orden de las tareas en la lista es relevante: siempre se insertan al final y se accede por posición.

🔄 Ejemplo de uso:

```
t1 = Task("Lavar los platos", 15)
t2 = Task("Hacer la compra", 40)
t3 = Task("Estudiar Python", 120)

lista = TaskList()
lista.add_task(t1)
lista.add_task(t2)
lista.add_task(t3)
print(lista) # ➡️[Lavar los platos: 15 min]->[Hacer la compra: 40 min]->[Estudiar Python: 120 min]->None

print(lista.get_task(1))       # Hacer la compra: 40 min
print(lista.time_left)         # 175

print(lista.task_completed(1)) # Hacer la compra: 40 min
print(lista) # ➡️[Lavar los platos: 15 min]->[Estudiar Python: 120 min]->None
print(lista.time_left)         # 135
print(lista.time_spent)        # 40
```

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Python 3 Cheat Sheet

Latest version on :
https://perso.limsi.fr/pointal/python:memento

Base Types

integer, float, boolean, string, bytes

```
int 783  0 -192  0b010 0o642 0xF3
float 9.23 0.0  -1.7e-6
bool True False
str "One\nTwo"
    'I\'m'
bytes b"toto\xfe\775"
```

zero *binary* *octal* *hexa*

escaped new line *escaped '* *escaped tab* *hexadecimal* *octal* *$\times 10^{-6}$*

Multiline string: `"""X\tY\tZ 1\t2\t3"""`

immutables

Container Types

- **ordered sequences**, fast index access, repeatable values

```
list [1,5,9]   ["x",11,8.9]   ["mot"]   []
tuple (1,5,9)  11,"y",7.4     ("mot",)  ()
str bytes
```

Non modifiable values (immutables) *expression with only comas → `tuple`* *(ordered sequences of chars / bytes)*

- **key containers**, no *a priori* order, fast key access, each key is unique

dictionary `dict {"key":"value"}` `dict(a=3,b=4,k="v")` `{}`
(key/value associations) `{1:"one",3:"three",2:"two",3.14:"π"}`

collection `set {"key1","key2"}` `{1,9,3,0}` `set()`

keys=hashable values (base types, immutables…) `frozenset` *immutable set* *empty*

Identifiers

for variables, functions, modules, classes… names

`a…zA…Z_` followed by `a…zA…Z_0…9`
- diacritics allowed but should be avoided
- language keywords forbidden
- lower/UPPER case discrimination

☺ `a toto x7 y_max BigOne`
☹ `~~By and for~~`

Variables assignment

`=`

assignment ⇔ **binding** of a *name* with a *value*
1) *evaluation of right side expression value*
2) *assignment in order with left side names*

```
x=1.2+8+sin(y)
a=b=c=0      assignment to same value
y,z,r=9.2,-7.6,0   multiple assignments
a,b=b,a      values swap
a,*b=seq     unpacking of sequence in
*a,b=seq     item and list
x+=3         increment ⇔ x=x+3
x-=2         decrement ⇔ x=x-2
x=None       « undefined » constant value
del x        remove name x
```

and `*=` `/=` `%=` …

Conversions

`type(expression)`

```
int("15") → 15
int("3f",16) → 63          can specify integer number base in 2nd parameter
int(15.56) → 15            truncate decimal part
float("-11.24e8") → -1124000000.0
round(15.56,1)→15.6        rounding to 1 decimal (0 decimal → integer number)
bool(x)  False for null x, empty container x , None or False x ; True for other x
str(x)→ "…"   representation string of x for display (cf. formatting on the back)
chr(64)→'@'  ord('@')→64   code ↔ char
repr(x)→ "…"  literal representation string of x
bytes([72,9,64]) → b'H\t@'
list("abc") → ['a','b','c']
dict([(3,"three"),(1,"one")]) → {1:'one',3:'three'}
set(["one","two"]) → {'one','two'}
separator str and sequence of str → assembled str
  ':'.join(['toto','12','pswd']) → 'toto:12:pswd'
str splitted on whitespaces → list of str
  "words with  spaces".split() → ['words','with','spaces']
str splitted on separator str → list of str
  "1,4,8,2".split(",") → ['1','4','8','2']
sequence of one type → list of another type (via list comprehension)
  [int(x) for x in ('1','29','-3')] → [1,29,-3]
```

Sequence Containers Indexing

for lists, tuples, strings, bytes…

negative index	−5	−4	−3	−2	−1
positive index	0	1	2	3	4
`lst=[10,`	`20,`	`30,`	`40,`	`50]`	
positive slice	0	1	2	3	4	5
negative slice	−5	−4	−3	−2	−1	

Items count
`len(lst)→5`

index from 0
(here from 0 to 4)

Individual access to **items** via `lst[index]`
`lst[0]→10` ⇒ *first one* `lst[1]→20`
`lst[-1]→50` ⇒ *last one* `lst[-2]→40`

On mutable sequences (`list`), remove with `del lst[3]` and modify with assignment `lst[4]=25`

Access to **sub-sequences** via `lst[start slice:end slice:step]`

```
lst[:-1]→[10,20,30,40]   lst[::-1]→[50,40,30,20,10]   lst[1:3]→[20,30]   lst[:3]→[10,20,30]
lst[1:-1]→[20,30,40]     lst[::-2]→[50,30,10]         lst[-3:-1]→[30,40] lst[3:]→[40,50]
lst[::2]→[10,30,50]      lst[:]→[10,20,30,40,50] shallow copy of sequence
```

Missing slice indication → from start / up to end.
On mutable sequences (`list`), remove with `del lst[3:5]` and modify with assignment `lst[1:4]=[15,25]`

Boolean Logic

Comparisons : `< > <= >= == !=` (*boolean results*) ≤ ≥ = ≠

`a and b` logical and *both simultaneously*

`a or b` logical or *one or other or both*

pitfall : `and` *and* `or` *return value of* `a` *or of* `b` *(under shortcut evaluation).*
⇒ *ensure that* `a` *and* `b` *are booleans.*

`not a` logical not

`True` `False` True and False constants

Statements Blocks

parent statement :
statement block 1…
⋮
parent statement :
statement block2…
⋮
next statement after block 1

indentation

configure editor to insert 4 spaces in place of an indentation tab.

Modules/Names Imports

module `truc`⇔file `truc.py`

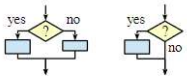
```
from monmod import nom1,nom2 as fct
  →direct access to names, renaming with as
import monmod →access via monmod.nom1 …
```

modules and packages searched in *python path* (cf `sys.path`)

Conditional Statement

statement block executed only if a condition is true

```
if logical condition :
    statements block
```



Can go with several *elif*, *elif*… and only one final *else*. Only the block of first true condition is executed.

```
if age<=18:
  state="Kid"
elif age>65:
  state="Retired"
else:
  state="Active"
```

with a var `x`:
`if bool(x)==True:` ⇔ `if x:`
`if bool(x)==False:` ⇔ `if not x:`

Maths

floating numbers… approximated values *angles in radians*

Operators: `+ - * / // % **`
Priority `(…)` × ÷ ↑ ↑ a^b integer ÷ ÷ remainder

`@` → matrix × *python3.5+`numpy`*

```
(1+5.3)*2→12.6
abs(-3.2)→3.2
round(3.57,1)→3.6
pow(4,3)→64.0
```

usual order of operations

```
from math import sin,pi…
sin(pi/4)→0.707…
cos(2*pi/3)→-0.4999…
sqrt(81)→9.0  √
log(e**2)→2.0
ceil(12.5)→13
floor(12.5)→12
```

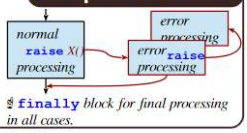
modules `math`, `statistics`, `random`, `decimal`, `fractions`, `numpy`, *etc. (cf. doc)*

Exceptions on Errors

Signaling an error:
`raise` *ExcClass(…)*

Errors processing:

```
try:
    normal procesising block
except Exception as e:
    error processing block
```



`finally` *block for final processing in all cases.*

Conditional Loop Statement

statements block executed ***as long as*** *condition is true*

```
while logical condition:
    statements block
```

```
s = 0
i = 1
while i <= 100:
    s = s + i**2
    i = i + 1
print("sum:",s)
```

initializations ***before*** *the loop condition with a least one variable value (here* `i`*)*

make condition variable change !

beware of infinite loops!

Algo: $s = \sum_{i=1}^{i=100} i^2$

Loop Control

`break`	*immediate exit*
`continue`	*next iteration*

`else` *block for* ***normal*** *loop exit.*

Display

```
print("v=",3,"cm :",x,",",y+4)
```

items to display : literal values, variables, expressions

`print` options:

- `sep=" "` — items separator, default space
- `end="\n"` — end of print, default new line
- `file=sys.stdout` — print to file, default standard output

Input

```
s = input("Instructions:")
```

`input` always returns a **string**, convert it to required type (cf. boxed *Conversions* on the other side).

Generic Operations on Containers

`len(c)` → items count
`min(c)` `max(c)` `sum(c)`
`sorted(c)` → `list` sorted *copy*
`val in c` → boolean, membership operator `in` (absence `not in`)
`enumerate(c)` → *iterator* on (index, value)
`zip(c1,c2...)` → *iterator* on tuples containing c_i items at same index
`all(c)` → `True` if **all** `c` items evaluated to true, else `False`
`any(c)` → `True` if **at least one** item of `c` evaluated true, else `False`

Note: For dictionaries and sets, these operations use ***keys****.*

Specific to ***ordered sequences containers*** *(lists, tuples, strings, bytes...)*

`reversed(c)` → *inversed iterator* `c*5`→ duplicate `c+c2`→ concatenate
`c.index(val)` → *position* `c.count(val)` → *events count*

```
import copy
copy.copy(c) → shallow copy of container
copy.deepcopy(c) → deep copy of container
```

Operations on Lists

modify original list

`lst.append(val)`	add item at end
`lst.extend(seq)`	add sequence of items at end
`lst.insert(idx, val)`	insert item at index
`lst.remove(val)`	remove first item with value *val*
`lst.pop(/idx/)` → *value*	remove & return item at index *idx* (default last)
`lst.sort()` `lst.reverse()`	sort / reverse liste *in place*

Operations on Dictionaries

`d[key]=value` `d.clear()`
`d[key]` → *value* `del d[key]`
`d.update(d2)` { update/add associations
`d.keys()` `d.values()` `d.items()` } → *iterable views on keys/values/associations*
`d.pop(key[,default])` → *value*
`d.popitem()` → *(key,value)*
`d.get(key[,default])` → *value*
`d.setdefault(key[,default])` → *value*

Operations on Sets

Operators:

- `|` → union (vertical bar char)
- `&` → intersection
- `-` `^` → difference/symmetric diff.
- `<` `<=` `>` `>=` → inclusion relations

Operators also exist as methods.

`s.update(s2)` `s.copy()`
`s.add(key)` `s.remove(key)`
`s.discard(key)` `s.clear()`
`s.pop()`

Files

storing data on disk, and reading it back

```
f = open("file.txt","w",encoding="utf8")
```

- **variable** for operations
- **name** of file on disk (+path...)
- opening **mode**: `'r'` read, `'w'` write, `'a'` append, `...'+' 'x' 'b' 't'`
- **encoding** of chars for *text files*: `utf8` `ascii` `latin1` ...

cf. modules `os`, `os.path` and `pathlib`

writing

`f.write("coucou")`
`f.writelines(list of lines)`

reading

read empty string if end of file

`f.read([n])` → *next chars* — *if n not specified, read up to end !*
`f.readline()` → *next line*
`f.readline()` → `list` *of next lines*

text mode `t` *by default (read/write* `str`*), possible binary mode* `b` *(read/write* `bytes`*). Convert from/to required type !*

`f.close()` — dont forget to **close the file** after use !

`f.flush()` write cache `f.truncate([size])` resize

reading/writing progress sequentially in the file, modifiable with:

`f.tell()`→*position* `f.seek(position[,origin])`

Very common: opening with a guarded block (automatic closing) and reading loop on lines of a text file:

```
with open(...) as f:
    for line in f :
        # processing of line
```

Iterative Loop Statement

statements block executed ***for each*** *item of a container or iterator*

```
for var in sequence:
    statements block
```

Go over sequence's **values**

```
s = "Some text"
cnt = 0
for c in s:
    if c == "e":
        cnt = cnt + 1
print("found",cnt,"'e'")
```

initializations ***before*** *the loop*

loop variable, ***assignment managed by*** `for` *statement*

Algo: count number of `e` *in the string.*

loop on dict/set ⇔ loop on keys sequences
use *slices* to loop on a subset of a sequence

Go over sequence's **index**

- modify item at index
- access items around index (before / after)

```
lst = [11,18,9,12,23,4,17]
lost = []
for idx in range(len(lst)):
    val = lst[idx]
    if val > 15:
        lost.append(val)
        lst[idx] = 15
print("modif:",lst,"-lost:",lost)
```

Algo: limit values greater than 15, memorizing of lost values.

good habit : don't modify loop variable

Go simultaneously over sequence's **index** and **values**:

```
for idx,val in enumerate(lst):
```

Integer Sequences

`range([start,] end [,step])`

`start` default 0, *end* not included in sequence, *step* signed, default 1

`range(5)` → 0 1 2 3 4 `range(2,12,3)` → 2 5 8 11
`range(3,8)` → 3 4 5 6 7 `range(20,5,-5)` → 20 15 10
`range(len(seq))` → *sequence of index of values in seq*

range provides an immutable sequence of int constructed as needed

Function Definition

function name (identifier)
named parameters

```
def fct(x,y,z):
    """documentation"""
    # statements block, res computation, etc.
    return res
```

result value of the call, if no computed result to return: `return None`

parameters and all variables of this block exist only *in* the block and *during* the function call (think of a "black box")

Advanced: `def fct(x,y,z,*args,a=3,b=5,**kwargs):`

**args variable positional arguments (→*`tuple`*), default values,*
***kwargs variable named arguments (→*`dict`*)*

Function Call

```
r = fct(3,i+2,2*i)
```

storage/use of returned value — *one argument per parameter*

this is the use of function name *with parentheses* which does the call

Advanced: **sequence* ***dict*

Operations on Strings

`s.startswith(prefix[,start[,end]])` `s.endswith(suffix[,start[,end]])` `s.strip([chars])`
`s.count(sub[,start[,end]])` `s.partition(sep)` → *(before,sep,after)*
`s.index(sub[,start[,end]])` `s.find(sub[,start[,end]])`
`s.is...()` *tests on chars categories (ex.* `s.isalpha()`*)*
`s.upper()` `s.lower()` `s.title()` `s.swapcase()`
`s.casefold()` `s.capitalize()` `s.center([width,fill])`
`s.ljust([width,fill])` `s.rjust([width,fill])` `s.zfill([width])`
`s.encode(encoding)` `s.split([sep])` `s.join(seq)`

Formatting

formating directives — values to format

```
"modele{} {} {}".format(x,y,r) → str
```

`"{selection : formatting ! conversion}"`

- **Selection** :
  `2`
  `nom`
  `0.nom`
  `4[key]`
  `0[2]`

Examples:

```
"{:+2.3f}".format(45.72793)
→'+45.728'
"{1:>10s}".format(8,"toto")
→'      toto'
"{x!r}".format(x="I'm")
→'"I\'m"'
```

- **Formatting** :

fill char *alignment* *sign* *mini width* . *precision~maxwidth* *type*

`< > ^ =` `+ - space` `0` at start for filling with 0

integer: `b` binary, `c` char, `d` decimal (default), `o` octal, `x` or `X` hexa...
float: `e` or `E` exponential, `f` or `F` fixed point, `g` or `G` appropriate (default),
string: `s` ... `%` percent

- **Conversion** : `s` (readable text) or `r` (literal representation)

Ficheros requeridos

prueba.py

```python
from tasks import TaskList
from tasks import Task

t1 = Task("Lavar los platos", 15)
t2 = Task("Hacer la compra", 40)
t3 = Task("Estudiar Python", 120)

lista = TaskList()
lista.add_task(t1)
lista.add_task(t2)
lista.add_task(t3)
print(lista) # ➡️[Lavar los platos: 15 min]->[Hacer la compra: 40 min]->[Estudiar Python: 120 min]->None

print(lista.get_task(1))       # Hacer la compra: 40 min
print(lista.time_left)         # 175

print(lista.task_completed(1)) # Hacer la compra: 40 min
print(lista) # ➡️[Lavar los platos: 15 min]->[Estudiar Python: 120 min]->None
print(lista.time_left)         # 135
print(lista.time_spent)        # 40
```

tasks.py

```python
from containerbase import ContainerBase

class Task:

    def __init__(self, name, duration):
        self.name = name
        self.duration = duration

    def __str__(self):
        return f"{self.name}: {self.duration} min"


class TaskList(ContainerBase):

    class Node:
        def __init__(self, value, next=None):
            self.value = value
            self.next = next

    def __init__(self):
        self.first = None
        self.last = None

```